

UTrucking AI Phone Assistant — QA & Testing Log

Prepared: 2026-07-02 **Agent:** Utrucking Agent (Retell AI) · versions v29 → v33 **How tested:** scripted conversations run against the *live* agent via Retell's playground API, one real phone call, and direct probes of the lookup backend + Google Sheets.

1. Summary

The assistant was tested across **three layers**: (1) does it behave correctly turn-by-turn, (2) does it hold up to hard/ambiguous names at scale, and (3) does it work on a real call. Headline results:

- **10 / 10 functional scenarios passed** (order lookup, privacy gate, disambiguation, FAQ, transfer, call-end, identity verification).
 - **Hard-name stress test:** a garbled name **never** resolved to the wrong student (0 wrong matches across the sample).
 - **1 live phone call** completed successfully end-to-end; one minor wording tic was found and fixed.
 - **1 privacy risk** in the backend (over-eager name matching) was identified **and mitigated** by an added identity-verification step.
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2. Functional / behavior tests

Each scenario was driven as a real conversation against the published agent.

#	Scenario	Expected	Result
1	Caller gives name, checks order	Confirms identity, then answers	✓ Pass
2	One-question-at-a-time answers	Answers only the field asked, briefly	✓ Pass

#	Scenario	Expected	Result
3	Order mentioned <i>before</i> a name is given	Asks for the name first, no premature lookup	✓ Pass (fixed in v30)
4	Privacy gate — wrong/close name match	Confirms the name; reveals nothing if "that's not me"	✓ Pass
5	Ambiguous name → multiple matches	Offers choices, lets caller pick	✓ Pass
6	General question (pricing/services)	Brief, accurate answer from knowledge base	✓ Pass
7	Caller says goodbye	Warm close, then ends the call	✓ Pass
8	Caller asks for a person	Connects to the UTrucking team (transfer)	✓ Pass
9	Name not found after spelling	Escalates to the team instead of looping	✓ Pass (fixed in v32)
10	Identity verification	Confirms a second detail; if wrong, does not share and transfers	✓ Pass (added in v33)

3. Live phone-call test

A real call was placed to the assistant (caller: a real student record). The assistant fuzzy-matched a mispronounced name, confirmed identity, and answered pickup location, status, order ID, billing, delivery, and website questions — each concisely. **Finding:** the assistant occasionally tacked on filler ("right?", "is that okay?"). **Action:** fixed in v32. Full transcript: `utrucking-test-calls.txt`.

4. Name-matching stress test

Automated audit against the live roster (~1,655 students).

A. Misspelled hard-to-pronounce names (18 tested) — a letter-swap was applied to real names to simulate speech-to-text errors: - **10** matched to the correct student exactly - **8** returned "let me confirm which one" (the assistant then disambiguates) - **0** matched the **wrong** student - **0** failed to find anything

B. Fake names not in the system (12 tested) — should never match a real student: - **10** correctly rejected - **2** were over-matched to a real student by the backend (~17%) - **Mitigation:** the v33 identity-verification step blocks these — the caller cannot confirm a stranger's building, so no data is shared. **Root fix:** tighten the backend match threshold (planned).

5. Integration / infrastructure checks

Check	Result
Backend reachable (<code>/lookup_student</code> , <code>/health</code> , <code>/debug_sheets</code>)	✅ Online
Dispatch Google Sheet	✅ ~1,655 rows, all expected columns present
Service Google Sheet	⚠️ Returns 0 rows (empty or un-shared) — item-list/invoice answers unavailable
Retell tools wired (<code>lookup_student</code> , <code>transfer_to_office</code> , <code>end_call</code>)	✅ Verified
Guardrails (jailbreak/abuse protection)	✅ Enabled (v33)

6. Known issues & mitigations

1. **Backend over-matching** (fake name → real student): mitigated by identity verification; permanent fix = raise the match threshold in the backend and redeploy.
 2. **Service sheet empty:** item/invoice details unavailable until the second Google Sheet is restored.
 3. **No phone number provisioned yet:** tested via the Retell dashboard/API; a live line can be added when ready.
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7. Method note

Behavior was validated by replaying full conversations against the **live agent** (not a mock) and inspecting every assistant message and tool call. Name matching was audited directly against

the production backend and Google Sheets. Testing spanned agent versions v29 through v33; each fix was re-tested before publishing.